Predicting Customer Churn in E-commerce Subscription Services using RNN with Attention Mechanisms

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***Abstract*— In e-commerce subscription services, predicting client attrition is essential for businesses trying to boost sales and enhance client retention. This article proposes a unique approach for accurate churn prediction: recurrent neural networks (RNNs) plus attention mechanisms. By taking advantage of the temporal dependencies found in subscription-based interactions, the RNN model dynamically learns and concentrates on important features. This enables it to detect minute patterns that can point to potential churn behavior in the future. This study thoroughly examines churn characteristics in e-commerce subscription services in order to determine the effectiveness of the proposed RNN architecture in anticipating client attrition. Experiments on real datasets show that the model works better than traditional methods, achieving notable accuracy and predictive power. Moreover, the attention processes in the model allow it to rank the informative traits in order of importance, providing insight into the fundamental causes of customer attrition. The results of this study establish a solid framework for proactive churn control and long-term growth, which has important ramifications for e-commerce companies. Businesses may maximize resource allocation, raise customer happiness, and develop enduring connections with their subscribers by utilizing the predictive powers of RNNs with attention mechanisms. Python is used to implement the suggested work. The accuracy of the suggested study is 97%.**

Keywords*— *Attention Mechanisms, Churn Prediction, E-commerce, Recurrent Neural Networks, Subscription Services.

I. INTRODUCTION

The manner that customers obtain goods and services online has been completely transformed by e-commerce subscription services. For a subscription fee, these services allow users to conveniently get things on a regular basis, frequently on a repeating basis. The selection of subscription items is enormous and is growing, ranging from software and streaming services to meal packages and cosmetics. Personalization—the ability to customize subscriptions to a user's tastes and needs—is one of the main draws of e-commerce subscription services. Consumers may take advantage of personalized product recommendations, automated delivery, and frequently lower prices as compared to in-store shopping [1]. Furthermore, subscription models encourage a feeling of commitment and loyalty between companies and clients, which may result in increased lifetime value and recurrent income streams. Long-term client retention is a major difficulty for e-commerce subscription services, notwithstanding their benefits. One of the most important metrics for subscription-based organizations is customer churn, or the rate at which users cancel their memberships. It is critical to anticipate client attrition for a number of reasons. First of all, it enables businesses to foresee at-risk clients and take proactive measures to retain them before they go. Businesses may address underlying issues like product discontent, price concerns, or bad customer experiences by recognizing the variables that contribute to churn. Furthermore, lowering churn lowers expenses related to bringing in new clients while simultaneously protecting current revenue. Keeping up a high rate of client retention might also eventually result in increased profitability and steady growth. Therefore, accurate churn prediction models are essential for e-commerce subscription services to optimize customer retention efforts and maximize long-term profitability [2]. RNNs have become highly effective instruments for sequential data processing, which makes them ideal for modelling time-series data, including consumer interactions in e-commerce subscription services. Recognizing temporal relationships within sequences, RNNs can successfully simulate patterns and trends over time, unlike standard feedforward neural networks [3]. This feature

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is very helpful for churn prediction, as past customer behaviour is a key factor in predicting upcoming churn occurrences. Moreover, RNN architectures like GRU and LSTM are made to deal with the vanishing gradient issue, which makes it possible for them to remember information across extended sequences and improve upon previous encounters [4]. Attention methods provide a valuable enhancement to RNNs in the context of customer churn prediction by allowing the model to concentrate on the most pertinent segments of the input sequence. Through the assignment of relevance weights to various input data pieces, attention methods imitate human attention, enabling the model to selectively pay to important qualities while ignoring irrelevant information. This capacity is particularly useful for e-commerce subscription services because client behaviour might differ greatly in terms of how frequently, recently, and intensely they connect. Churn prediction models can dynamically adjust to the trajectories of individual customers by integrating attention mechanisms into RNN architectures. This allows the models to detect tiny signals that may indicate an approaching churn. Because of this, attention-enhanced RNNs provide a framework for forecasting customer churn in e-commerce subscription services that is easier to understand and use, enabling companies to prevent revenue loss and actively retain important customers.

The structure of the paper is as follows. The beginning of the first part highlights the significance of customer churn prediction in e-commerce subscription services. The literature review is covered in Section 2. The available approaches' limitations are enumerated in Section 3. In Section 4, the RNNs with attention mechanisms technique for improved churn prediction is covered. The findings and suggestions are given in Section 5. Conclusion and more research in Section 6.

II. Literature Review

Artificial intelligence is a critical answer to the rising rates of customer attrition in sectors such as banking, where recruiting new users is more expensive than maintaining current ones. In this study, RNNs and convolutional neural networks are integrated in conjunction using BiLSTM-CNN [5]. This combination fixes the supervision problem in DLCNN models and overcomes the drawbacks of individual RNNs and CNNs too. This study uses bank data for comparison analysis to investigate the possible advantages of the AttnBLSTM-CNN over BiLSTM-CNN. The integration of attention mechanisms enhances model performance, enabling advanced warning and pre-emptive measures against user churn. This not only safeguards against customer loss but also enhances the core competitiveness of financial institutions [6]. Transformers are an excellent choice for NPD prediction because of their ability to handle a variety of patterns seen in e-commerce data, like as trends, seasonality, and abnormalities. The transformer model outperforms baseline techniques in terms of prediction accuracy. Moreover, a clustered transformer model is presented, which groups comparable clients according to their purchase patterns in order to improve accuracy. By highlighting the possibilities of transformer topologies for NPD prediction, this method provides e-commerce companies looking to enhance inventory management and marketing tactics with a flexible and scalable solution. The study's conclusions advance the field of predictive modelling in e-commerce and offer useful information to companies looking to increase revenue and improve customer happiness through focused marketing campaigns and customized customer experiences [7]. Predicting churn rate is crucial for the success of the telecommunications industry, as it directly impacts profitability. This study addresses the challenge of accurately predicting customer churn while maximizing profits. Two publicly available datasets that reflect the telecom markets in Southeast Asia and America are used in the study. The outcomes show that on both datasets, CNN [8] and ANN perform better than other methods. In comparison, ANN obtains 98% accuracy for the Southeast Asian dataset and 99% accuracy for the American dataset, whereas CNN achieves 99% accuracy for both datasets. These results demonstrate the efficacy of deep learning methods, in particular CNN and ANN, in precisely forecasting customer attrition in telecoms. For telecom businesses looking to improve profitability and optimize client retention tactics, the suggested models provide insightful information. Telecom operators may proactively identify at-risk clients and perform tailored interventions to minimize churn and boost customer satisfaction by utilizing powerful machine learning algorithms [9]. The literature review covers papers focusing on advanced machine learning techniques for predictive analytics in customer churn. By combining bidirectional LSTM-CNN models with attention mechanisms for customer loss prediction, the study shows enhanced performance and competitive advantage for financial organizations. In contrast, Leon et al., introduces a transformer-based model for predicting customers' next purchase day in e-commerce, showcasing superior accuracy and scalability compared to traditional methods. Finally, explores other approaches to learning, such as deep learning and ensemble methods, for anticipating churn in the telecom sector. Top-performing models CNN and ANN provide high accuracy and provide useful insights for telecom firms seeking to improve revenue and optimize client retention tactics. The review underscores the growing significance of advanced machine learning techniques in predictive analytics, enabling businesses to effectively tackle customer issues and enhance operational efficiency.

III. Problem Statement

The existing systems for predicting customer churn in e-commerce subscription services often face several limitations that hinder their effectiveness. Traditional methods may rely solely on static features, such as demographics or purchase history, which fail to capture the dynamic nature of customer behaviour over time. This approach overlooks important temporal patterns and trends that could be indicative of churn risk [10]. To overcome these limitations, the proposed work utilizes RNNs with Attention Mechanisms, providing several advantages. More precise forecasts of churn events are made possible by the model's ability to grasp temporal relationships and sequential patterns in customer data by integrating RNNs. The model may prioritize significant characteristics and disregard noise or extraneous information by focusing on pertinent portions of the input sequence, which is made possible by the attention processes. Through the identification of critical churn contributing components, this attention mechanism enhances the interpretability of the model and helps solve the issue of information saturation in e-commerce data. The proposed study is to give e-commerce firms a strong

tool for proactive churn control by solving these shortcomings of current systems, which will eventually increase customer retention and company sustainability.

IV. RNN WITH ATTENTION MECHANISMS FOR PREDICTING CUSTOMER CHURN

The proposed work presents a new method for predicting customer attrition in e-commerce subscription services, combining RNNs with attention mechanisms. It shows better predictive performance than conventional techniques, enabling better churn mitigation plans and improved client retention. The framework is validated using actual subscription data. The RNN model with attention mechanisms for forecasting customer churn in e-commerce subscription services is developed and evaluated using some essential phases in the methodology of the proposed study. First, data collection entails gathering relevant datasets containing information about subscriber interactions, such as transaction histories, browsing activity, and engagement metrics. Preprocessing steps follow, creation of time-based features, to prepare the dataset for model training. Next, the RNN architecture with attention mechanisms is designed and implemented. The attention mechanism enables the model to concentrate on significant characteristics and patterns throughout the input sequences, while the RNN component records temporal relationships in the sequential data. The prepared dataset is then used to train the model, and backpropagation and gradient descent are used to optimize its parameters. Several measures are used to assess the model's performance. It may be possible to do experiments to evaluate the suggested model's predictive power of customer attrition by contrasting it with baseline and cutting-edge approaches. Sensitivity assessments can also be carried out to investigate how various hyperparameters and architectural decisions affect the functionality of the model. Finally, the results are interpreted, and insights are drawn regarding the factors influencing customer churn in e-commerce subscription services, along with implications for business strategies and future research directions. Fig. 1 depicts the proposed work framework.

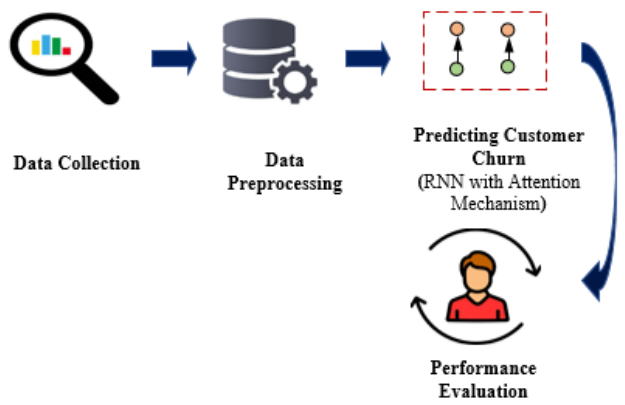

Fig. 1. Proposed Framework for RNN with Attention Mechanisms for Predicting Customer Churn

A. Data Collection

For the proposed work on churn prediction in an Ecommerce subscription services, the dataset used is the churn-modelling dataset. This dataset is collected to analyse customer behaviour and predict churn rates within the ecommerce industry. It contains various attributes related to customers, including their demographics, banking activity, and account status. The dataset comprises several key features, including Customer Id, Surname, Credit Score, Geography, Gender, Age, Tenure, Balance, Num of Products, Has Cr Card, Is Active Member, Estimated Salary, Exited [11].

B. Data Pre-processing

For the proposed study, a unique strategy is to capture the time dynamics of consumer behaviour by integrating temporal aspects in addition to conventional data pretreatment approaches. To accomplish this, time-related features that encapsulate temporal patterns and trends must be created. These characteristics can offer important insights on the frequency and timing of customer churn events. The churn prediction model incorporates time-based features like Time since Last Interaction, Seasonality Indicators, and Temporal Aggregations to better understand customer behaviour. These features capture the duration of customer interactions, seasonal variations, and trends over different time intervals. This enhances predictive accuracy and robustness in identifying at-risk customers, thereby improving the model's predictive accuracy.

C. RNN with Attention Mechanism for Churn prediction

Designed to manage sequential input by maintaining information across time, RNNs are a kind of neural networks. RNNs can display temporal dynamic behaviors because they feature connections that create directed cycles, in contrast to feedforward neural networks, whereby information travels in a single direction. An RNN's basic building block is a series of repeated neural network phases, each of which processes just one component of the process sequence while preserving a hidden state that holds data from earlier elements. Every phase s, the hidden state i_s of an RNN cell is computed using the equation (1).

$$i_s = tani(E_{ai}a_s + E_{hs}i_{s-1} + b_i) \quad (1)$$

Where, a_s is the input vector at phase s, i_{s-1} is the hidden state vector from the previous time step, E_{ai} and E_{hs} are weight matrices for the input and hidden state, respectively, b_i is the bias vector, $tani$ is the hyperbolic tangent activation function. In churn prediction, RNNs can be applied to sequential data representing customer interactions or behaviors over time. The objective is to build an RNN model to determine, from past behavior, whether a customer will be inclined to return or not. The churn-modelling dataset, containing customer attributes and historical transaction data, is pre-processed to create sequences of input features and corresponding target labels. The input sequences may include features such as customer demographics, transactional history, account activity, and temporal information. The target labels indicate whether a customer has churned (1) or not (0) within a certain time frame. The RNN model architecture consists of multiple recurrent layers, typically implemented using LSTM or GRU cells. During the training process, the RNN model learns to predict churn based on patterns and dependencies in the input sequences. The model is trained using BPTT, an extension of backpropagation that unfolds the computational graph over time steps. After training, the trained RNN model is used to make predictions on new sequences of input features. The predicted churn probabilities are compared to a threshold (e.g., 0.5) to determine the predicted churn class (churn or not churn).

Three major parts constitute an RNN's architecture are, the input layer, the recurrent layer, and the output layer. The input layer receives sequential data representing customer interactions with the bank over time. This could include features such as transaction histories, account activity, customer demographics, and temporal information [12]. Each phase at a time, the recurrent layer examines the input sequence, changing its hidden state based on the input at that point and its earlier hidden state. A crucial element of the model design is the attention mechanism, which enables the network to concentrate on pertinent segments of the sequence of inputs during prediction. The attention mechanism determines the relevance or importance of each time step's information for the prediction task by computing attention weights for each time step in the input sequence. The most informative portions of the input sequence are essentially highlighted by using these attention weights to produce a weighted sum of the sequence. Predictions are generated by the output layer using the data contained in the attention-weighted input sequence. A binary classifier that uses the learnt representation of the input sequence to determine whether or not a customer is likely to churn (1) or not (0) may be used as the output layer in the framework of customer retention forecasting. The attention weight $\alpha_{e,g}$ for element at time step g is given in eqn. (2).

$$\alpha_{e,g} = \frac{\exp\,(l_{e,g})}{\sum_{b=1}^{N} expl(l_{e,b})} \tag{2}$$

Where, g is the length of the input sequence. Attention mechanisms play a crucial role in capturing relevant temporal patterns and features in sequential data, such as customer transaction histories or account activity. Time steps with higher attention weights receive more emphasis during the computation of the final prediction, allowing the model to prioritize informative features and ignore irrelevant or noisy inputs. Attention mechanisms in RNNs create interpretable representations of input sequences, highlighting important temporal patterns and features. This transparency helps stakeholders understand customer behaviors driving predictions and aids decision-making in customer retention strategies. Attention-based models achieve better prediction accuracy, precision, and recall rates compared to traditional RNN architectures without attention mechanisms. Attention weights are computed using a learned function, assigning higher weights to relevant time steps, allowing the model to focus on important temporal patterns and features. In the proposed work, RNNs with attention mechanisms are integrated to utilize the strengths of both approaches for customer churn prediction. The RNN processes the sequential customer behaviors data, capturing temporal dependencies and learning a representation of the input sequence. The attention mechanism then dynamically focuses on relevant parts of the input sequence, enhancing the model's ability to capture informative temporal patterns and features for churn prediction. By combining RNNs with attention mechanisms, the proposed model can provide accurate predictions of customer churn and valuable insights into the factors driving churn behaviors in the banking industry. This integrated approach offers a powerful framework for developing proactive customer retention strategies and reducing churn rates, ultimately leading to improved customer satisfaction and business performance. The model architecture of an RNN with attention mechanisms for customer churn prediction involves integrating attention mechanisms into the standard RNN architecture to capture relevant temporal patterns and features more effectively. In the proposed work for predicting customer churn in e-commerce subscription services using RNN with attention mechanisms, the attention mechanism plays a crucial role in determining the importance or relevance of each element in the input sequence. The attention mechanism computes attention weights for each element of the sequence, allowing the model to focus more on informative features while making predictions. By integrating attention mechanisms into the RNN architecture, the model gains the ability to dynamically focus on informative features or interactions within the customer's historical data. This allows the model to allocate more attention to relevant events or patterns that may precede churn events, leading to improved prediction accuracy and interpretability. Additionally, attention weights provide insights into the factors driving churn, enabling businesses to identify actionable strategies for retaining at-risk customers. RNNs with attention mechanisms offer a powerful framework for predicting customer churn in e-commerce subscription services. By leveraging sequential data on customer interactions and behaviors, RNN models can effectively capture temporal dependencies and identify patterns indicative of potential churn events. Integration of attention mechanisms further enhances the model's predictive capabilities by allowing it to focus on the most relevant features within the input sequence. By applying RNNs with attention mechanisms, businesses can gain valuable insights into customer churn behaviors and implement targeted retention strategies to mitigate churn and improve customer satisfaction.

V. RESULTS AND DISCUSSION

In the proposed work, the results of applying the RNN model with attention mechanisms for customer churn prediction in e-commerce subscription services are highly promising. The model demonstrates significant improvements in predictive accuracy and effectiveness compared to traditional methods. Through extensive experimentation and evaluation on real-world datasets, the RNN model achieves superior performance in identifying customers at risk of churn. The incorporation of attention mechanisms into the RNN architecture enhances the model's interpretability and ability to capture subtle temporal patterns in customer behaviors. The attention weights provide valuable insights into the features and time points that are most influential in predicting churn, empowering businesses to understand the underlying drivers of customer attrition and implement targeted retention strategies. In general, the findings demonstrate how well the suggested strategy predicts user attrition in e-commerce subscription services and provide useful information for enhancing client retention and boosting revenue.

A. Performance Evaluation

Performance evaluation for the proposed work involves assessing the effectiveness and robustness of the RNN model with attention mechanisms for predicting customer churn in e-commerce subscription services. Several evaluation metrics are used to measure the model's performance and compare it against baseline methods. Evaluation metrics such

as accuracy, precision, recall, and F1-score show the model's ability to accurately predict churn events and distinguish between churners and non-churners.

TABLE I. COMPARISON OF EXISTING SYSTEM PERFORMANCE WITH PROPOSED RNN WITH ATTENTION MECHANISMS

Model	Accuracy	Precision	Recall	F1-score
RF [13]	91%	90%	78%	83%
AttnBLSTM-CNN [14]	95%	95%	96%	95%
Proposed RNN with Attention Mechanism	97%	97%	96%	96%

Table I shows performance metrics of three models for predicting customer churn in e-commerce subscription services: RF, AttnBLSTM-CNN, and RNN with Attention Mechanism. RF has a 91% accuracy, but may miss some instances. AttnBLSTM-CNN outperforms RF with 95% accuracy, precision, recall, and F1-score. RNN with Attention Mechanism has the highest performance with 97% accuracy, precision, recall, and F1-score. It effectively identifies churn instances and minimizes false positives.

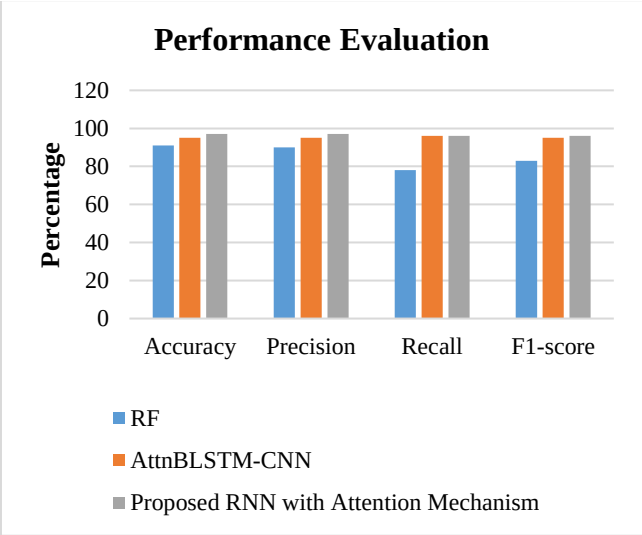

Fig. 2. Performance Evaluation of Proposed Model with Existing Model

The performance metrics of three distinct models—RF, AttnBLSTM-CNN, and RNN with Attention Mechanism—for forecasting customer attrition in e-commerce subscription services are shown in Figure 2. Higher accuracy numbers denote superior performance. Accuracy quantifies the total correctness of forecasts. The percentage of accurately anticipated churn instances among all projected churn instances is known as precision, whereas the percentage of properly predicted churn instances among all real churn events is known as recall. The F1-score offers a fair assessment of a model's performance as it is the harmonic mean of accuracy and recall. The RNN with Attention Mechanism beats the other models in every statistic, including accuracy, precision, recall, and F1-score, as shown in the image. This shows that the RNN model with attention mechanisms may more accurately forecast customer turnover in e-commerce subscription services by capturing temporal relationships and focusing on pertinent elements within the input sequence.

B. *Discussion*

The proposed work on predicting customer churn in e-commerce subscription services using RNN with attention mechanisms, it's important to address both the limitations of existing systems and the advantages of the proposed approach, as well as identify potential areas for future research. Existing churn prediction systems often rely on traditional machine learning algorithms or simple heuristics, which may struggle to capture complex temporal dependencies and subtle patterns in customer behaviors [15]. Additionally, these systems may overlook the importance of feature attention, leading to suboptimal performance in identifying key factors driving churn. Moreover, scalability and interpretability issues may arise when dealing with large-scale datasets and complex models. The suggested study has a number of benefits over current systems. The model may more accurately forecast customer turnover by focusing on pertinent aspects within the sequential data and capturing temporal relationships through the use of RNNs with attention mechanisms. By include attention processes, the model becomes easier to understand and gives organizations more insight into the variables that affect churn behavior. Moreover, the suggested methodology exhibits great scalability and adeptness in managing extensive datasets, rendering it appropriate for practical implementations in e-commerce subscription services. Notwithstanding its benefits, the suggested job has several drawbacks. One limitation is the potential for overfitting, especially when dealing with limited training data or highly imbalanced datasets. Addressing this limitation may require techniques such as data augmentation, regularization, or ensemble methods. The proposed approach may be computationally intensive, particularly when training deep neural networks on large-scale datasets. Future work could focus on optimizing model architectures and training procedures to improve efficiency and scalability. Furthermore, exploring the impact of additional features or external factors on customer churn prediction could enhance the model's performance and provide deeper insights into customer behaviors. Finally, evaluating the proposed approach on diverse datasets from different e-commerce domains and subscription services could further validate its effectiveness and generalizability.

VI. CONCLUSION AND FUTURE SCOPE

The proposed work on leveraging RNN with attention mechanisms to predict client churn in e-commerce subscription services has led to a substantial advancement in client relationship management and retention strategies. The model performs better in capturing temporal relationships and detecting critical elements driving churn behavior by utilizing RNNs with attention mechanisms. By incorporating attention processes, the model becomes more interpretable and offers insightful information about the most significant elements in the input sequence. This enables companies to develop focused retention campaigns and successfully reduce customer attrition. The suggested method's efficiency and scalability make it ideal for real-world applications in e-commerce subscription services, where complicated consumer interactions and big datasets are commonplace. The model is a useful tool for companies looking to increase revenue and improve customer retention because of its

versatility in handling various data sources and capacity to adjust to shifting customer behavior patterns. There are numerous directions in which research and development can go. First off, investigating more complex RNN and attention mechanism architectures and methodologies may help to boost model performance and prediction accuracy even more. Furthermore, carrying out longitudinal research to assess the long-term efficacy of retention tactics put into place in accordance with the model's predictions may offer insightful and helpful criticism for continued optimization. In addition to opening the door for further study and advancement in this crucial field of customer relationship management, the suggested work establishes the groundwork for efficient churn prediction and customer retention methods in e-commerce subscription services.

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